

HOMEWORK 1

Pedestrian Detection

From Traditional Method (HOG + SVM)
to Deep Learning (ANN & Transfer Learning)

521H0324 – Nguyen Van Truong

521H0047 – Le Nguyen Nhat Hao

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Chương 1

Experimental Results and Discussion

This chapter presents the experimental findings obtained from the proposed pedestrian detection pipeline. Three approaches are investigated, including the traditional **HOG + SVM** method, a simple **Artificial Neural Network (ANN)**, and a deep learning approach based on **MobileNetV2 Transfer Learning**. The comparison focuses on detection capability, classification performance, computational efficiency, and robustness under challenging scenarios.

1.1 HOG + SVM Benchmark Results

The first experimental stage evaluated the **Histogram of Oriented Gradients (HOG)** combined with a **Support Vector Machine (SVM)** detector for pedestrian detection.

The HOG+SVM model was initially tested on four homework images (`IMG_1.jpg` to `IMG_4.jpg`) using the baseline configuration:

$$\text{winStride} = (4, 4)$$

$$\text{scale} = 1.05$$

Under this setting, the detector identified 6, 1, 5, and 0 pedestrians, respectively.

The corresponding inference times were approximately 0.4383s, 0.1110s, 0.0796s, and 0.2384s, which correspond to frame rates of approximately 2.28, 9.01, 12.56, and 4.19 FPS.

To investigate the impact of parameter tuning, benchmark experiments were performed using multiple configurations of `winStride` and `scale`.

Specifically:

$$\text{winStride} \in \{(2, 2), (4, 4), (8, 8)\}$$

and

$$\text{scale} \in \{1.03, 1.05\}$$

The experimental results extracted directly from the notebook are summarized in Table 1.1.

Bảng 1.1: Benchmark results of HOG+SVM on homework images

Image	winStride	Scale	Detected	Time (s)	FPS
IMG_1.jpg	(2,2)	1.03	5	1.2067	0.83
IMG_1.jpg	(2,2)	1.05	5	0.7343	1.36
IMG_1.jpg	(4,4)	1.03	5	0.2979	3.36
IMG_1.jpg	(4,4)	1.05	5	0.1853	5.40
IMG_1.jpg	(8,8)	1.03	5	0.0863	11.59
IMG_1.jpg	(8,8)	1.05	4	0.0547	18.27
IMG_2.jpg	(2,2)	1.03	2	0.7434	1.35
IMG_2.jpg	(2,2)	1.05	1	0.4716	2.12
IMG_2.jpg	(4,4)	1.03	1	0.1997	5.01
IMG_2.jpg	(4,4)	1.05	1	0.1274	7.85
IMG_2.jpg	(8,8)	1.03	1	0.0578	17.31
IMG_2.jpg	(8,8)	1.05	0	0.0375	26.63
IMG_3.jpg	(2,2)	1.03	5	0.4627	2.16
IMG_3.jpg	(2,2)	1.05	5	0.2776	3.60
IMG_3.jpg	(4,4)	1.03	5	0.1236	8.09
IMG_3.jpg	(4,4)	1.05	5	0.0815	12.27
IMG_3.jpg	(8,8)	1.03	5	0.0358	27.92
IMG_3.jpg	(8,8)	1.05	5	0.0262	38.13
IMG_4.jpg	(2,2)	1.03	2	0.5390	1.86
IMG_4.jpg	(2,2)	1.05	0	0.3309	3.02

The benchmark results demonstrate a clear trade-off between *detection accuracy* and *computational speed*.

For example, smaller **winStride** values generally improved pedestrian detection performance but significantly reduced processing speed. In `IMG_1.jpg`, using

$$\text{winStride} = (8, 8)$$

achieved approximately 18.27 FPS but detected only 4 pedestrians.

In contrast, using

$$\text{winStride} = (2, 2)$$

detected 5 pedestrians, but reduced the frame rate to only 0.83 FPS.

The highest computational efficiency was observed on `IMG_3.jpg` using

$$\text{winStride} = (8, 8)$$

and

$$\text{scale} = 1.05$$

which achieved an inference time of only

$$0.0262s$$

equivalent to approximately

$$38.13 \text{ FPS}$$

while successfully detecting all five pedestrians.

However, this aggressive parameter configuration may increase the likelihood of missed detections in more challenging scenarios. For example, in `IMG_1.jpg`, the detector identified only 4 pedestrians under this configuration, compared with 5 detections using

$$\text{winStride} = (4, 4)$$

and

$$\text{scale} = 1.05.$$

Overall, the experiments clearly indicate that smaller stride values improve detection quality at the expense of computational speed, highlighting the classical trade-off between accuracy and efficiency in traditional computer vision systems.

1.2 Simple Artificial Neural Network (ANN)

To establish a deep learning baseline, a simple **Artificial Neural Network (ANN)** was trained for binary pedestrian classification.

The dataset was constructed using the **COCO 2017 validation split**, where images were filtered into two categories:

- **PERSON**
- **NON-PERSON**

To maintain dataset balance, 1000 samples were selected for each class, resulting in a total dataset of 2000 images.

All images were resized into a fixed resolution of

$$64 \times 128 \times 3$$

and normalized into the range

$$[0, 1]$$

by dividing pixel values by 255.0.

The dataset was split into training, validation, and testing subsets, consisting of:

- Training set: 1400 images
- Validation set: 300 images
- Testing set: 300 images

1.2.1 ANN Architecture

The ANN model was intentionally designed to be lightweight and computationally simple.

The architecture is summarized as follows:

Flatten $\rightarrow$ Dense(512, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(1, Sig

The **Flatten** operation converts a three-dimensional image tensor into a one-dimensional vector before feeding the data into fully connected layers.

Although computationally simple, this operation removes **spatial information** contained within the image, such as local texture, shape structure, and neighborhood relationships between pixels.

As a consequence, the model becomes significantly less effective for visual recognition tasks, particularly pedestrian detection, which relies heavily on spatial and structural features.

1.2.2 Training Results

The ANN model was trained using EarlyStopping, which terminated training after six epochs due to limited performance improvement.

The final training performance reached approximately:

$$\text{Training Accuracy} \approx 52.9\%$$

and

$$\text{Validation Accuracy} \approx 49.3\%$$

with corresponding losses of approximately

$$0.70$$

and

$$0.69.$$

The testing performance was evaluated using the held-out test set.

The final evaluation results were:

$$\text{Test Accuracy} \approx 54.67\%$$

and

$$\text{Test Loss} \approx 0.6899.$$

These results indicate that the ANN model performed only slightly better than random guessing, which would be expected to achieve approximately

$$50\%$$

accuracy in a balanced binary classification task.

1.2.3 Performance Analysis

Several factors contributed to the weak performance of the ANN model.

First, the flattening process destroyed the spatial structure of images, preventing the network from learning meaningful regional features such as body contours, human posture, and texture patterns.

Second, the network architecture remained relatively shallow, consisting of only two hidden dense layers. This limited representational capacity reduced the ability of the model to learn high-level abstractions from image data.

Third, the experimental training curves observed in the notebook indicate that both training and validation performance stabilized quickly after only a few epochs. This behavior suggests that the model suffered from **underfitting**, meaning it lacked sufficient capacity to capture the complexity of pedestrian appearance.

Overall, the ANN baseline demonstrated poor classification capability and proved insufficient for reliable pedestrian recognition tasks. These findings highlight the importance of preserving spatial information when designing deep learning architectures for computer vision.

1.3 Transfer Learning with MobileNetV2

To improve pedestrian classification performance, a transfer learning strategy was employed using **MobileNetV2**, a lightweight convolutional neural network pretrained on the **ImageNet** dataset.

Unlike the simple ANN baseline, MobileNetV2 preserves spatial information through convolutional operations, making it significantly more suitable for visual recognition tasks.

1.3.1 MobileNetV2 Architecture

The pretrained MobileNetV2 backbone was loaded using

```
include_top=False
```

which removes the original fully connected classification layers trained for ImageNet classes.

The convolutional feature extractor (backbone) was retained, while a custom classifier head was added to adapt the network to the pedestrian classification problem.

The final classification architecture can be summarized as:

MobileNetV2 Backbone → GlobalAveragePooling → Dense → Dropout → Sigmoid

The backbone layers were almost entirely frozen, with approximately

99%

of pretrained weights kept unchanged. Only the newly added classifier head was trained during optimization.

This strategy significantly reduced training cost while preserving powerful visual representations learned from ImageNet.

1.3.2 Why MobileNetV2 Performs Better

The superior performance of MobileNetV2 can be explained by its architecture.

First, the model utilizes **depthwise separable convolutions**, which decompose standard convolution into two simpler operations:

- Depthwise convolution (feature extraction per channel)
- Pointwise convolution (feature combination)

This design dramatically reduces computational complexity while maintaining strong representational power.

Second, MobileNetV2 employs **inverted residual blocks** with **bottleneck layers**, allowing efficient feature reuse and better gradient propagation.

Finally, transfer learning from ImageNet provides the model with strong low-level and mid-level visual features such as edges, textures, object shapes, and semantic patterns, which are highly beneficial for pedestrian classification tasks.

Consequently, the model can generalize effectively even with a relatively limited training dataset.

1.3.3 Training Results

The MobileNetV2 model was trained using the same training, validation, and testing sets used for the ANN baseline.

Training was terminated automatically using EarlyStopping at approximately epoch seven.

The final training performance achieved:

Training Accuracy $\approx 98.4\%$

and

Validation Accuracy $\approx 81.7\%$.

Final testing performance was reported as:

Test Accuracy $\approx 82.00\%$

and

Test Loss ≈ 0.3854 .

Compared to the ANN baseline, MobileNetV2 improved classification accuracy by approximately

27%.

This large performance gap demonstrates the effectiveness of convolutional feature extraction and transfer learning for computer vision applications.

1.3.4 Performance Analysis

The strong performance of MobileNetV2 suggests that pretrained convolutional networks can effectively learn meaningful pedestrian representations, even under relatively constrained training conditions.

However, the training results also reveal a noticeable performance gap between training and validation accuracy:

98.4% vs. 81.7%.

This difference may indicate a mild degree of **overfitting**, where the model learns training patterns more effectively than unseen validation data.

Despite this limitation, the testing accuracy of approximately 82% remains substantially higher than the ANN baseline.

Overall, MobileNetV2 achieved the best trade-off between robustness, classification performance, and practical usability among all evaluated approaches.

1.4 Homework Image Prediction

To evaluate the generalization capability of the trained model, the MobileNetV2 classifier was tested on four homework images:

- IMG_1.jpg
- IMG_2.jpg
- IMG_3.jpg

- IMG_4.jpg

Figure 1.1 illustrates the four homework images used for evaluation.



(a) IMG_1



(b) IMG_2



(c) IMG_3



(d) IMG_4

Hình 1.1: Homework input images used for evaluation

The classification threshold was set to

0.15.

Experimental observations indicate that all images containing pedestrians were correctly classified as

PERSON

with high confidence scores.

For example, IMG_1.jpg and IMG_3.jpg were successfully identified as pedestrian-containing images.

Meanwhile, images without pedestrians, such as IMG_4.jpg, were classified as

NON-PERSON

with low confidence values.

Unlike MobileNetV2, the ANN model was not explicitly used for inference on the homework images, as only MobileNetV2 predictions were visualized in the notebook.

These findings further emphasize the superior generalization capability of MobileNetV2, while the ANN baseline remained close to random classification behavior.

1.5 Comparative Analysis

A comprehensive comparison between the evaluated approaches is summarized in Table 1.2.

Bảng 1.2: Comparison of pedestrian detection approaches

Criteria	HOG+SVM	ANN	MobileNetV2
Accuracy	Moderate	Low	High
Inference Speed	High	High	Moderate
Spatial Awareness	Moderate	Poor	Strong
Occlusion Handling	Weak	Weak	Better
Viewpoint Robustness	Limited	Weak	Strong
Small Pedestrian Handling	Weak	Poor	Better
Training Required	No	Yes	Yes
Computational Cost	Low	Moderate	High
Real-Time Potential	High	Moderate	High (GPU)

Experimental findings reveal that MobileNetV2 achieved the strongest overall performance among all methods.

Although HOG+SVM provided fast inference speed and demonstrated strong real-time potential, its performance was highly dependent on parameter tuning and scene conditions.

The ANN baseline, despite its simplicity, performed poorly due to the loss of spatial information caused by image flattening.

MobileNetV2 successfully balanced classification accuracy, robustness, and practical applicability, making it the most effective approach for pedestrian recognition in this study.

1.6 Failure Analysis

Although the evaluated approaches demonstrated promising performance, several limitations remain.

1.6.1 HOG + SVM Limitations

The HOG+SVM detector is highly sensitive to:

- Non-standard viewpoints (e.g., top-down perspective)
- Small pedestrian sizes
- Occlusion scenarios
- Complex backgrounds

The default detector primarily focuses on upright human body structures. As a result, pedestrians with unusual poses or partially hidden appearances are more likely to be missed.

In addition, parameter selection presents a significant challenge.

Larger stride values improve speed but increase the risk of missed detections, while smaller strides improve detection quality at the expense of inference time.

1.6.2 ANN Limitations

The ANN baseline failed primarily because the **Flatten** operation destroyed spatial image relationships.

As a consequence, the network could not effectively learn important visual patterns such as:

- Human body shape
- Object boundaries
- Texture patterns
- Local spatial context

Additionally, the shallow architecture and relatively small dataset restricted feature learning capability, resulting in clear underfitting behavior.

This explains why the ANN model performed only marginally better than random guessing.

1.6.3 MobileNetV2 Limitations

Although MobileNetV2 achieved substantially stronger performance, several limitations still exist.

First, the gap between training and validation accuracy suggests a mild degree of overfitting.

Second, very small pedestrian regions may still remain difficult to classify accurately, particularly due to the relatively small input resolution of

$$64 \times 128.$$

Finally, the model may experience **domain shift** between the COCO dataset and real-world homework images, particularly when lighting conditions, camera viewpoints, or pedestrian appearances differ significantly from training examples.

1.7 Conclusion

This study investigated pedestrian detection using both traditional computer vision and deep learning approaches.

The HOG+SVM method demonstrated strong computational efficiency and remained suitable for lightweight real-time applications. However, its performance was highly dependent on parameter tuning and scene conditions.

The ANN baseline showed poor performance, achieving approximately 54.67% testing accuracy, which was only slightly higher than random guessing. This limitation was primarily caused by the destruction of spatial information through image flattening.

In contrast, MobileNetV2 Transfer Learning achieved the best performance, reaching approximately

$$82\%$$

testing accuracy.

The strong performance of MobileNetV2 highlights the importance of convolutional feature extraction and transfer learning in computer vision applications.

Future improvements may include:

- Dataset augmentation
- Fine-tuning additional backbone layers
- Custom CNN architectures
- Hybrid detection pipelines

- More diverse pedestrian datasets

Overall, deep learning approaches, particularly transfer learning, proved substantially more effective than traditional methods for pedestrian recognition tasks.

Tài liệu tham khảo

- [1] Navneet Dalal and Bill Triggs, *Histograms of Oriented Gradients for Human Detection*, IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2005.
- [2] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen, *MobileNetV2: Inverted Residuals and Linear Bottlenecks*, CVPR, 2018.
- [3] TensorFlow Documentation, *TensorFlow and Keras API*, Available at: <https://www.tensorflow.org>
- [4] COCO Dataset, *Common Objects in Context*, Available at: <https://cocodataset.org>